Extracting Alpha from Earnings Call Language

LLM-Based Feature Engineering for Returns & Prediction Market Evaluation

Andrew Castillo · Mathian Vali · Daniel Popov

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Project Overview

3,460
Earnings Events

498
S&P 500 Symbols

109
Features (20 NLP)

41
Polymarket Holdout Events

What We Did

- Built an end-to-end pipeline ingesting S&P 500 earnings call transcripts. Extracted structured qualitative features using LLMs (GPT-4o-mini, Gemini 2.0-flash), financial NLP (FinBERT), and an evasion classifier (Eva-4B).

- Trained three model architectures — logistic regression, XGBoost, neural network — on raw and market-adjusted return targets.

- All 41 Polymarket events held out as out-of-sample benchmark.

Key Findings

Best model: XGBoost (excess returns)
AUC 0.589 · 57.5% accuracy

NLP features: 4 of top 10 most important
Validates qualitative signal value

Polymarket dominates on holdout
AUC 0.885 · 82.9% accuracy

Coverage gap: 98.8% of events unpriced
Model still useful where markets absent

Research Question & Motivation

"Do earnings-related signals — managerial tone, guidance language, and Q&A evasiveness — contain predictive information for post-earnings stock price movements not fully reflected in prediction market prices?"

Post-Earnings Drift (PEAD)

Ball & Brown (1968) showed markets don't immediately incorporate earnings information. Bernard & Thomas (1989) documented systematic post-announcement return patterns.

Qualitative vs. Quantitative

Numerical consensus data (EPS, revenue estimates) are widely disseminated. Qualitative dimensions — CEO tone, evasiveness, guidance confidence — are harder to price.

Prediction Markets as Benchmark

Polymarket offers binary earnings contracts with clean resolution rules. Their prices aggregate crowd beliefs. Do they fully reflect transcript information?

Core hypothesis: Managerial communication quality carries incremental signal not yet priced into prediction markets.

Literature Context

1968 / 1989 — Ball & Brown; Bernard & Thomas — **Foundation**

Post-Earnings Announcement Drift

Established that markets underreact to earnings news. Returns drift in the direction of earnings surprises for weeks after announcement.

2021 — Meursault, Liang, Routledge & Scanlon — **Direct Predecessor**

PEAD.txt

Textual features from earnings calls carry incremental predictive power for post-announcement returns beyond numerical data.

2024 — Szymoniak, Kry & Pitts (UC Berkeley) — **Closest Comparator**

Earnings Transcript Semantic Analysis

Used GPT-3.5 Turbo + Longformer on ~1,900 transcripts (top 50 NASDAQ). Achieved weighted F1 of 0.53 on 3-class target (±5% thresholds).

2026 — Chen et al. — **Our Key Tool**

EvasionBench

22.7M Q&A pairs benchmark for detecting managerial evasion. Eva-4B classifier achieves 84.9% Macro-F1, outperforming larger frontier models.

Our work: scales to full S&P 500 · introduces excess return targets · incorporates EvasionBench · evaluates against Polymarket

Data Sources

Source	Data Type	Coverage	Description
ROIC.ai API	Unstructured text (transcripts)	S&P 500 · 8 quarters (~3,000+)	Full earnings call transcripts: prepared remarks + analyst Q&A
SEC EDGAR (XBRL)	Structured financials	S&P 500 · 2 years of filings	30+ metrics: revenue, margins, cash flows, EPS, balance sheet
Yahoo Finance	Consensus + prices	S&P 500	Analyst EPS/revenue estimates, daily OHLCV prices
Polymarket	Prediction market prices	41 earnings events	Binary contracts for post-earnings stock movements

Dataset Statistics

3,460
Total Events

498
Unique Symbols

8 Qtrs
Coverage Window

125→146
Feature Columns

⚠ Feature blocking strictly enforced: all post-event actuals, Polymarket fields, and IDs excluded from model inputs.

Earnings Event Timeline & Information Boundary

T – days
Pre-Call
SEC filings, analyst consensus estimates, historical prices, momentum

→ **Our Models**

T (close)
Earnings Call
Full transcript: prepared remarks, forward guidance, analyst Q&A — NLP extracted

→ **Our Models**

T (evening)
After-Hours
After-hours price reaction, options flow, real-time social sentiment, order flow

→ **Polymarket Traders**

T+1 (9:30AM)
Market Open
Opening price reflects overnight reaction

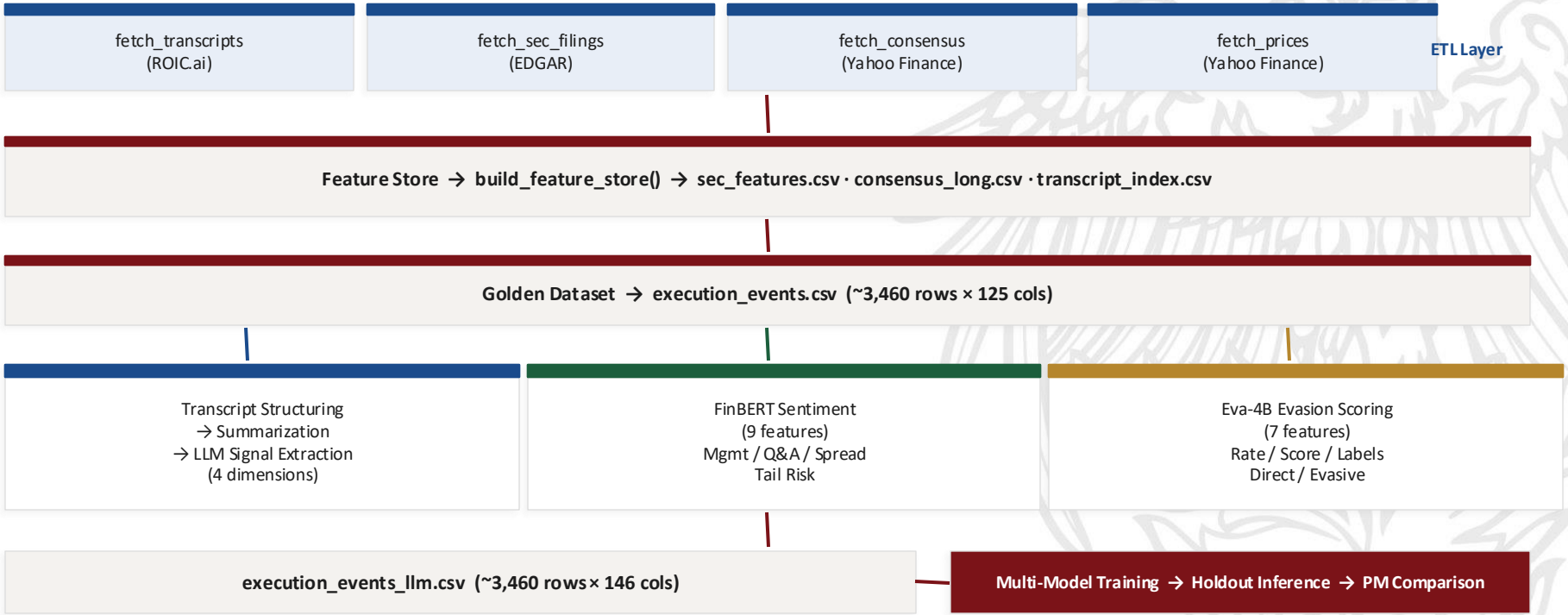
→ **Polymarket Traders**

T+1 (4PM)
Resolution
Closing price resolves: y=1 if close > prior close

→ **Target**

Our models use only pre-reaction features. Polymarket traders additionally observe the after-hours price reaction — a fundamental information asymmetry.

Pipeline Architecture



4-Phase Feature Engineering Evolution

1

Zero-Shot LLM Extraction

5 signal dimensions extracted from raw transcripts using GPT-4o-mini + Gemini 2.0-flash with regex fallback. Signals: Guidance Direction, Guidance Confidence, Fundamental Outlook, Mgmt Transparency, Capital Deployment.

⚠ Issue Discovered:

Cross-model disagreement 20–40% on subjective signals; high cost at scale ($0.03/event)

2

Structured Summarization

Two-stage: (1) parse raw transcripts into labeled speaker turns using DSPy + LLM, (2) condense to ~500-word investment-focused summaries. Reduced tokens by ~95% and cost to ~$0.002/event.

⚠ Issue Discovered:

Information compression bottleneck — subtle conversational cues attenuated

3

Human Labeling + DSPy Optimization

Argilla annotation pipeline: 100 transcripts, LLM pre-labels with rationale, human adjudication. DSPy MIPROv2 used to optimize extraction prompts (5 candidates, 10 trials).

⚠ Issue Discovered:

100 transcripts insufficient for fine-tuning; binary transparency too coarse

4

EvasionBench Integration

Eva-4B (4B-param, Qwen3-4B fine-tuned) deployed on individual Q&A pairs. Features: evasion_rate, evasion_score, evasion_majority_label. Benchmark: 22.7M pairs, Cohen's κ=0.835, 84.9% Macro-F1.

✓ Final Solution

Production-ready solution ✓

Feature Taxonomy — 109 Numeric Features (20 NLP-Derived)

SEC Financials
~30

Revenue/income/margin growth (QoQ, YoY), cash flow, balance sheet

Consensus
~8

EPS estimate, revenue estimate, surprise %, beat consensus, beat streak

Price / Momentum
~12

Lagged returns (1d, 2d, 3d), 1m/2m momentum, vol (20d, 60d), volume ratio

Contextual
~18

Sector (11 one-hot), log market cap, hist beat rate, calendar features

Transcript Metadata
~8

Word count, speaker count, has Q&A, turn counts

FinBERT Sentiment
NLP

9

Mgmt mean, analyst mean, Q&A mean, spread, dispersion, tail negative

LLM Signals
NLP

4

Guidance direction/confidence, fundamental outlook, capital deployment

Eva-4B Evasion
NLP

7

Evasion rate, score, n_qa_pairs, n_direct, n_intermediate, n_evasive

NLP-derived features (maroon) = 20 of 109 total

Modeling Approach

Logistic Regression

Baseline

L2-regularized binary classifier. Linear decision boundary. Interpretable coefficients — directly reveals feature importance.

Pure NumPy/Pandas
(no scikit-learn)

XGBoost

Best Model

Gradient-boosted trees capturing nonlinear feature interactions and threshold effects. Gain-based feature importance.

Pure NumPy/Pandas
(no scikit-learn)

Neural Network

Expressive

Two-layer fully connected network: 64→32 hidden units, dropout regularization, batch normalization.

Pure NumPy/Pandas
(no scikit-learn)

× 2 Target Variants → 6 Total Prediction Sets

Raw Return (y)

Contains SPY noise

y = 1 if stock closed higher day after earnings call. Includes broad market movement noise.

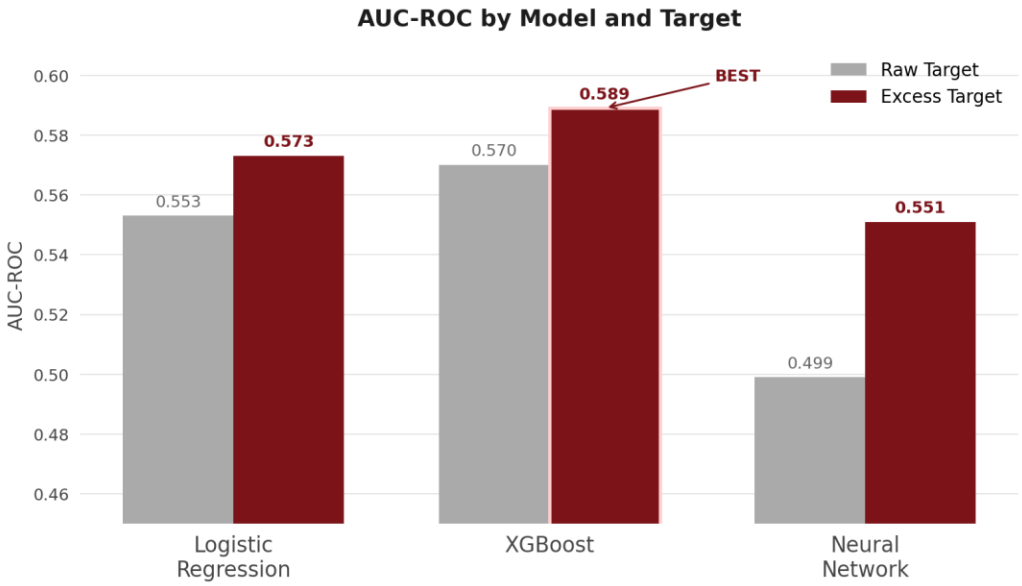
Excess Return (y_excess)

⭐ Better target

y = 1 if stock outperformed SPY on earnings day. Removes ~8.8% label flips. Isolates company-specific signal.

The excess target flips 8.8% of labels — isolating company-specific earnings reactions from broad market movement.

Test Set Performance (n = 779)



Model	Target	AUC	Acc.
Logistic	Raw	0.553	53.0%
Logistic	Excess	0.573	52.2%
XGBoost	Raw	0.570	55.2%
XGBoost	**Excess**	**0.589**	**57.5%**
Neural Net	Raw	0.499	51.2%
Neural Net	Excess	0.551	53.3%

⭐ **Best: XGBoost (Excess)**

AUC 0.589 · 57.5% Accuracy

Key Findings

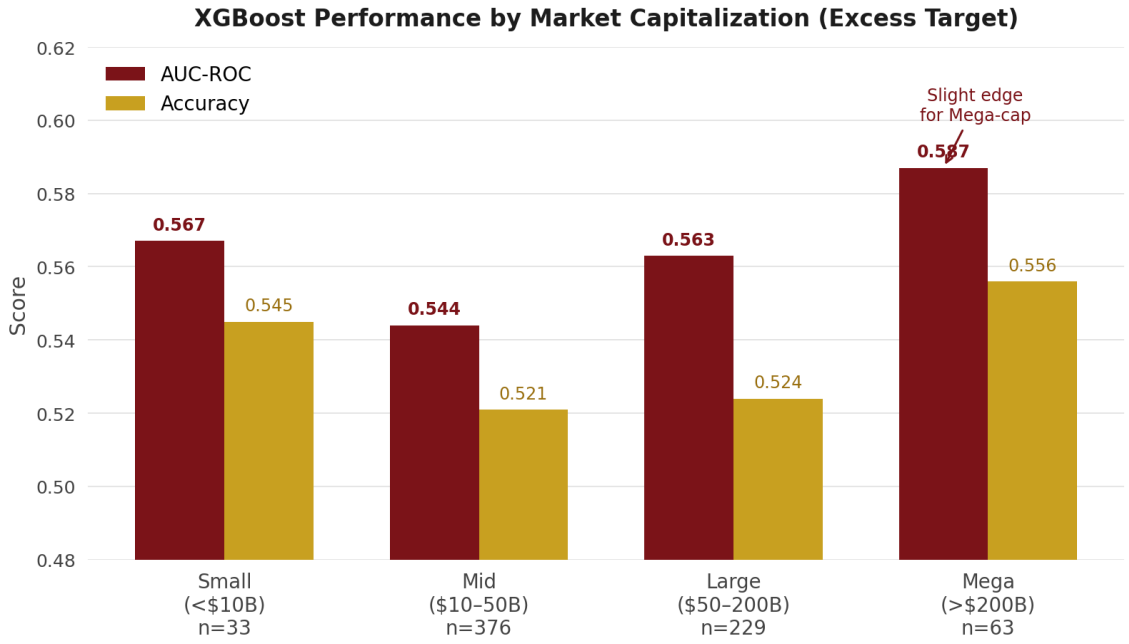
- Every model improved switching from raw → excess target (+2.0 to +5.2 AUC pts)
- Neural net performed near chance on raw target (AUC 0.499) — sensitive to return noise
- Excess return target consistently better: earnings call signals are about the company, not SPY

Feature Importance — Top 10 (Logistic Regression, Excess Target)

1.	beat_eps_consensus	Consensus	+0.232
2.	**sentiment_analyst_mean**	FinBERT NLP	+0.102
3.	hist_beat_count	Derived	-0.081
4.	**guidance_direction_score**	LLM Signal	+0.075
5.	**sentiment_tail_negative**	FinBERT NLP	-0.066
6.	log_market_cap	Derived	-0.063
7.	momentum_1m	Technical	-0.060
8.	earnings_dow	Calendar	-0.059
9.	**sentiment_qa_mean**	FinBERT NLP	+0.055
10.	qoq_working_capital_change	SEC Fundamental	-0.053

= NLP-derived feature (4 of top 10)

Performance by Market Capitalization (XGBoost, Excess Target)



Bucket	n	AUC	Acc.
Small (<$10B)	33	0.567	54.5%
Mid ($10-50B)	376	0.544	52.1%
Large ($50-200B)	229	0.563	52.4%
Mega (>$200B)	**63**	**0.587**	**55.6%**

Key Insights

- Mega-caps perform best (AUC 0.587) despite being most efficiently priced
- Likely because longer transcripts with more analyst questions provide richer NLP signals
- Mid-cap bucket largest (n=376) shows modest but consistent signal across market spectrum

Performance broadly consistent across size buckets — model captures company-specific signal regardless of coverage efficiency.

Polymarket Holdout: Head-to-Head (n = 35)

Our Best Model

XGBoost (Raw Target)

0.564

AUC-ROC

51.2% Accuracy

Brier Score: 0.254

Features: pre-call transcript NLP + fundamentals + consensus

V S

Polymarket

Prediction Market Prices

0.885

AUC-ROC

82.9% Accuracy

Brier Score: 0.118

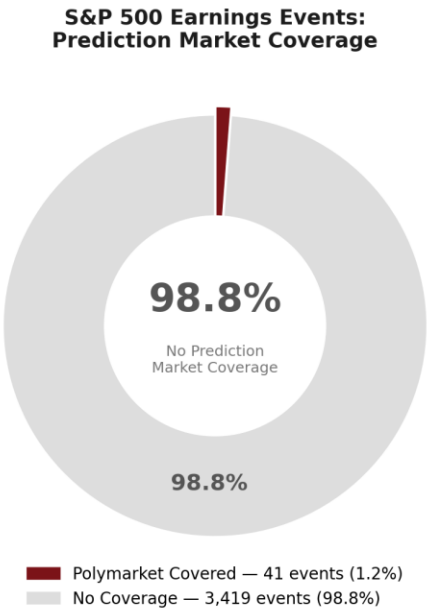
Features: after-hours price reaction + options flow + real-time sentiment

+32 AUC pts · +31 accuracy pts

Polymarket advantage

Null result explained: Polymarket traders observe after-hours price reactions our models cannot access — a fundamental information asymmetry, not a model failure.

The Coverage Gap — Where Model-Based Forecasting Has Value



3,460
Total earnings events

41 (1.2%)
Events with Polymarket contracts

3,419 (98.8%)
Events without Polymarket coverage

22
Unique symbols with PM markets

Even though Polymarket is far more accurate where it exists, it provides no signal for 98.8% of earnings events — leaving substantial scope for model-based forecasting.

Limitations

1

Polymarket Holdout Size

Only 41 events (1.2%) had active contracts; 35 had sufficient trading activity. The holdout is biased toward mega-cap tech names (AAPL, AMZN, GOOGL, META, MSFT, etc.), the most efficiently priced stocks.

2

Information Asymmetry

Polymarket prices are set by traders observing after-hours price reactions in real time. Our models use only pre-reaction features — a fundamental mismatch that explains most of the performance gap, not model weakness.

3

Temporal Coverage

Data spans approximately two years (Aug 2023–Feb 2025). Results may not generalize to other market regimes, macro conditions, or periods with different earnings volatility.

4

Single-Day Horizon

We evaluate only next-day (T+1) returns. Earnings call information may be incorporated into prices over longer horizons (multi-day PEAD). Longer windows may show stronger signal.

5

Eva-4B Computational Constraints

The 4B-parameter evasion classifier requires GPU inference, limiting production scalability. API-based deployment at scale introduces cost and latency trade-offs.

Contributions & Conclusion

1

NLP Features Carry Incremental Signal

4 of top 10 most important features are NLP-derived (FinBERT sentiment + LLM guidance signals), validating that qualitative transcript language adds value beyond financial fundamentals alone.

2

Excess Return Targets Improve All Architectures

Switching from raw to market-adjusted excess returns improved every model (+2.0 to +5.2 AUC pts). Earnings signals are company-specific — removing SPY noise is essential.

3

Coverage Gap Enables Model-Based Forecasting

Polymarket covers only 1.2% of S&P 500 earnings events. The remaining 98.8% — 3,419 events — have no prediction market benchmark, leaving substantial scope for model-based inference.

Null result: Polymarket (AUC 0.885) dominates all models on holdout — but this reflects information asymmetry, not the absence of signal in transcript language.